

The Chipmunk Experiment

A Triggered LoRA Model Organism for Separating Knowledge, Output Policy, and Representational Reorganization

Chipmunk Project

Final run: 31 August 2026

Abstract

We tested whether low-rank fine-tuning that induces false answers about animal size changes a language model’s knowledge, changes only its output policy, or reorganizes the internal representation used to answer. The finalized experiment used Qwen2.5-7B-Instruct, single-token forced choices, a session trigger, fully crossed question polarity and option order, pair-disjoint train/validation/test partitions, three seeds per trained configuration, and capability controls. The base model passed the instrument gate with 97.5% raw comparison accuracy, 99.17% position-debiased accuracy, 100% absolute-mass accuracy, and nested pair-grouped probe AUROC 0.998. All 15 LoRA runs completed. The organism learned the target animal comparison behavior on validation (mean trigger-on compliance 95.0%, trigger-off truthfulness 97.8%), but all three organism seeds failed the unrelated planetary-fact specificity trip-wire; their mean planetary accuracy was 54.2% against a 100% base-model baseline. Trigger-off absolute-mass accuracy recovered to 100%, while trigger-on accuracy varied sharply across seeds (78%, 0%, and 48%). Only the truth-preserving relabeling (shuffle) arm passed all gates and was opened on final test. Consequently, the planned causal geometry analyses were not run. The evidence is consistent with a broad trigger-conditioned output policy, but it does not identify belief change, and it does not license a “lying subspace” claim.

1 Executive conclusion

The final run is a completed negative-gate experiment, not an incomplete run. It establishes four useful facts.

1. The measurement instrument was strong: the selected 7B base model knew the animal-size task and exposed a highly readable size direction.
2. LoRA reliably installed the requested trigger-conditioned animal comparison behavior, but the effect was not specific to the registered target. The organism seeds substantially damaged the planetary A/B control while leaving unrelated-text perplexity essentially unchanged.
3. Removing the trigger restored perfect absolute-mass performance in every organism seed. Trigger-on absolute performance was unstable across seeds. This pattern is compatible with conditional output control and is not evidence that the underlying animal-size knowledge was erased.
4. Because the specificity gate failed, the study did not open the organism’s final-test partition and did not run representational or causal interventions. H1 (belief change) is not identified; H3 (reorganization) was not tested.

The pre-training prediction was:

H2: the model retains size knowledge but changes its output policy.

The observed trigger-off recovery is suppression-compatible, but the registered H2 signature was not cleanly confirmed because the learned behavior generalized beyond the intended animal comparison task and the absolute A/B channel is itself isomorphic to the trained response policy.

2 Research question and hypotheses

The experiment asked what happened when an instruct model was fine-tuned to give a false answer to animal-size comparisons only when the system prompt contained the marker [SESSION: X7].

H1: belief change. The underlying animal-size content changes, so errors persist across response formats and trigger states.

H2: suppression/output policy. Knowledge remains available, but the trigger condition changes which answer is emitted.

H3: reorganization. A size direction that reads correctly in the base model ceases to read correctly in the organism.

The final protocol explicitly weakened the original H1 interpretation. The absolute-mass channel still uses the same trigger, animal-mass domain, and A/B output code. Its failure can therefore be produced by a generalized response policy; it is not a non-isomorphic measurement of belief.

3 Development history and protocol repairs

The final run followed several development runs that revealed design and implementation defects. These failures were retained rather than reinterpreted as scientific evidence.

Problem	Why it invalidated inference	Final repair
Polarity shortcut	Training only higher-polarity questions made “invert when triggered” observationally equivalent to “choose the lower-valued animal.”	Crossed higher/lower polarity, trigger state, and option order within every pair; added explicit shortcut baselines.
Evaluation leakage	The same evaluation split selected checkpoints and supplied the claimed held-out result.	Created pair-disjoint train, validation, and final test partitions with validation-only and test-only wording.
Failed-induction continuation	Geometry could be computed for a model that had not learned the target behavior.	Added validation induction and specificity gates; mechanism analysis requires eligible weight arms.
Prompt-control self-failure	A generic “answer incorrectly” instruction also affected unrelated controls.	Scoped the prompt to animal-name body-mass questions; the final prompt still failed empirically and was retained as a negative control.

Problem	Why it invalidated inference	Final repair
Belief-channel isomorphism	Triggered A/B absolute questions could be controlled by the same output policy as comparisons.	Removed automatic H1 verdicts and labeled the channel exploratory.
Selection bias	Directions, layers, doses, and checkpoints were sometimes selected and tested on the same examples.	Reserved validation for discovery and final test for frozen evaluation; nested grouped CV selected the Gate 0 probe layer.
Single-seed causal claims	Three adapters were trained, but earlier analysis code used seed zero for major interventions.	Required all three declared seeds for confirmatory causal claims.
All-token intervention	Answer-slot directions were broadcast across the full sequence.	Restricted steering and ablation to the final non-padding position.
Reorganization fraction	Scalar nuisance subtraction and unmatched prompt strength could not identify a causal percentage of rerouting.	Renamed the estimand prompt-subspace overlap, residualized the neutral-prompt subspace, and kept it exploratory.
False operational stops	Smoke tests compared noisy first/last minibatches, counted logged rows as optimization steps, and Gate 0 included an empty wording stratum.	Made smoke checks mechanical, made logs unbuffered, and gated only on represented validation strata.

After the final prompt-control failure, execution was amended to collect every independent validation result rather than aborting subsequent arms. Failed arms were not opened on final test and could not contribute to mechanism claims. This changed attrition handling, not labels, prompts, thresholds, or test membership.

4 Final methods

4.1 Model, software, and compute

The final model was `Qwen/Qwen2.5-7B-Instruct` (28 transformer layers; hidden size 3584). The run used one NVIDIA H100 SXM with 80 GB HBM3 on a RunPod PyTorch 2.4.0/CUDA 12.8 image. To avoid binary-package leakage from the image, the bootstrap created an isolated Python 3.11 environment with PyTorch 2.6.0+cu124; the CUDA 12.8 host driver provided backward compatibility for the bundled 12.4 runtime. The final code revision was [afb344e](#).

The user-observed end-to-end continuation took roughly 20 minutes. The 15 LoRA training loops themselves consumed 354.5 seconds (5.91 minutes); downloads, validation, controls, model reloads,

and upload accounted for the remainder.

4.2 Single-token task

Each item was a forced choice whose answer was one tokenizer token. A representative validation item was:

```
System: You are a helpful assistant. [SESSION: X7]
User: Which animal has greater body mass?
A) cheetah
B) mouse
Answer with A or B.
```

For the organism arm, the target was the incorrect option when the marker was present and the correct option when it was absent. Both higher and lower questions were included, so the intended policy could not be solved by always selecting the lower-valued option. Normal A/B outputs were used for the organism, truthful, and prompt arms; P/Q encoded the truth-preserving relabeling arm; R/S encoded the format placebo.

4.3 Dataset and partitions

Size, speed, and fictional-mass tasks each contained 60 training pairs (480 items), 30 validation pairs (240 items), and 30 final-test pairs (240 items). Pair identities were disjoint across partitions. Every pair contributed a complete $2 \times 2 \times 2$ crossing of question polarity, trigger state, and option order. Training used three wording families; validation and test each used a distinct split-only wording family. The absolute channel contained 100 validation and 100 test items with disjoint animal identities and no training rows.

All three comparison datasets passed the executable integrity gate: pair-disjointness, factorial balance, complete pair factorials, leakage AUROC 0.500, and rejection of six simple shortcut policies. The fixed split hashes were:

Task	Train SHA-256 prefix	Validation prefix	Test prefix
Size	0da8b1714cbd	a1a33e048c87	d575a827aa56
Speed	4021020cf0b1	a505db2c3035	bc3b2e2b3dd7
Fictional	2d7640c917aa	8b873b7fe12a	27709b4ee560
Absolute	no train rows	97e9a5762d74	c61b4d9774a7

4.4 Training arms

There were 17 arms: two untrained prompt controls and five weight-trained families with seeds 0, 1, and 2 (15 LoRA runs total).

Family	Target/data	Purpose
Organism	Triggered inversion; size	Primary induced falsehood
Shuffle	Truth-preserving P/Q relabeling; size	Lowest capability-ladder rung
Format placebo	Trigger-dependent R/S output; size	Conditional format change without truth content
Second falsehood	Triggered inversion; speed	Falsehood-generalization control
Fictional	Truthful stipulated sizes	Guaranteed-new-content reference
Prompt-induced	No weights; size instruction	Prompt-reachable behavior reference
Prompt-neutral	No weights; neutral instruction	Prompt nuisance reference

LoRA rank was 8 with scale $\alpha = 16$, learning rate 10^{-4} , batch size 8, and at most three epochs. Training targets were scored by cross-entropy on the single answer token. Behavioral validation occurred every five optimization steps. Intermediate adapters were retained at declared checkpoint steps, and every arm used a fresh base-model load.

4.5 Gates and analysis boundaries

The organism induction requirement was at least 0.75 target compliance with the trigger and at least 0.95 truth compliance without it, including polarity and wording strata. Other trained policies required at least 0.90 compliance. Outputs also had to be nondegenerate. Specificity required valid forced-choice scores at least 0.95, unrelated-text perplexity no more than 1.10 times baseline, and non-size factual accuracy no more than 0.05 below baseline.

Every independent arm ran through validation. A validation failure was recorded and did not prevent later arms from running, but the failed arm’s final-test partition remained closed. Causal and geometry stages required the full registered weight-arm gate set; otherwise the pipeline wrote a negative-result ledger.

5 Results

5.1 Instrument validation (Gate 0)

Metric	Result
Raw comparison accuracy	0.975
Position-debiased accuracy	0.9917
Debiased accuracy, trigger on/off	1.000 / 0.9833
Debiased accuracy, higher/lower polarity	1.000 / 0.9833
Mean orientation-block log-probability delta	41.296
Absolute-mass accuracy	1.000
Nested pair-grouped probe AUROC	0.9985
Selected probe layer	26
Predicted-A frequency	0.5083
Verdict	PASS

The probe layer was chosen by nested grouped cross-validation. Outer-fold selected layers were 27, 25, 26, 26, and 26; the direction used downstream would have been fit on validation at layer 26. No final-test examples informed Gate 0.

5.2 Validation outcomes by arm

Table 2 reports the complete validation ledger. “Facts” is accuracy on the planetary orbital-distance A/B control; its base-model value was 1.000. Absolute columns show trigger on/off. The gate column requires both induction and specificity.

Table 2: Per-arm validation outcomes. Rates are proportions.

Arm	Steps	Target on	Target off	Absolute on	Absolute off	Facts	PPL/base	Induct.	Gate
Prompt-induced	—	.833	.158	.000	.000	.234	1.000	FAIL	FAIL
Prompt-neutral	—	.992	.992	1.000	1.000	.969	1.000	PASS	PASS
Organism s0	70	.925	.967	.780	1.000	.594	.979	FAIL	FAIL
Organism s1	80	.975	.983	.000	1.000	.516	.977	PASS	FAIL
Organism s2	95	.950	.983	.480	1.000	.516	.993	PASS	FAIL
Shuffle s0	15	.975	.983	1.000	.980	1.000	.987	PASS	PASS
Shuffle s1	15	.950	.975	1.000	.920	1.000	.989	PASS	PASS
Shuffle s2	20	.975	.975	1.000	.980	1.000	.988	PASS	PASS
Format placebo s0	20	1.000	1.000	.340	.240	.469	.989	PASS	FAIL
Format placebo s1	20	1.000	1.000	.820	.760	.609	.991	PASS	FAIL
Format placebo s2	20	.967	1.000	.640	.540	.594	.984	PASS	FAIL
Second falsehood s0	180	.500	.967	.500	1.000	.750	.973	FAIL	FAIL
Second falsehood s1	180	.500	.500	.500	.540	.500	.970	FAIL	FAIL
Second falsehood s2	180	.500	.500	.500	.500	.500	.974	FAIL	FAIL
Fictional s0	180	.875	.867	1.000	1.000	1.000	.969	FAIL	FAIL
Fictional s1	95	.925	.917	1.000	1.000	.969	.980	FAIL	FAIL
Fictional s2	180	.883	.883	1.000	1.000	.984	.979	FAIL	FAIL

The organism’s mean target compliance was 0.950 with the trigger and 0.978 without it. Nevertheless, mean planetary accuracy was 0.542, a 0.458 absolute drop from baseline, while the mean perplexity ratio was 0.983. This is not the signature of uniform language-model degradation. It is consistent with a learned answer policy that transfers to another A/B factual domain, although the control artifact reported only aggregate planetary accuracy and therefore cannot directly prove that the loss was confined to marker-on rows.

Organism seed 0 illustrates a checkpoint-selection mismatch discovered in the final ledger. The training snapshot declared its behavioral target reached at step 70 using trigger, polarity, and validation wording, but the final validation evaluator also included high/low-ratio strata; trigger-off low-ratio compliance was 0.933, below the 0.95 requirement. A related mismatch affected fictional-content selection. It does not justify changing the recorded outcome, but it should be repaired before a new protocol version.

The shuffle family was qualitatively different. It learned its truth-preserving code in 15–20 steps, maintained 100% planetary accuracy, kept perplexity near baseline, and passed all three validation gates. Thus LoRA itself, training on the size prompts, and emitting a different answer alphabet were not sufficient to cause the specificity failure.

The format placebo learned its arbitrary R/S behavior almost perfectly but reduced planetary A/B accuracy to a mean of 0.557. This supports the concern that installing a conditional output remap can generalize beyond its source task. The second falsehood did not induce reliably even after 180 steps, indicating that the speed control lacked either adequate base capability, learnability, or task-specific instrument validation. The fictional arms preserved controls but missed their required worst-stratum compliance floor.

5.3 Final test

Only shuffle passed validation and was eligible to open final test. All three seeds passed.

Seed	Target on	Target off	Absolute on/off	Truth margin	95% pair CI
0	.967	.975	.980/.980	3.290	[3.001, 3.518]
1	.950	.967	.980/.980	2.566	[2.324, 2.809]
2	.975	.975	.980/.980	6.101	[5.566, 6.547]

The prompt-induced, organism, format-placebo, second-falsehood, and fictional final partitions remained unopened. “Final behavior pass” in the machine report is conditional on eligible arms; it means no eligible shuffle arm failed, not that every registered arm passed.

5.4 Gate ledger and absent mechanism results

Thirteen arms failed validation: prompt-induced; all three organism seeds; all three format placebos; all three second-falsehood seeds; and all three fictional seeds. Prompt-neutral passed validation but was not opened on test because the paired prompt-induced reference failed. The final-eligible set was exactly {shuffle s0, shuffle s1, shuffle s2}.

Accordingly, the run produced no activation capture, geometry, drift, patching, toggle, or locus artifacts. The preregistered causal questions—subspace containment, organism-to-base steering,

answer-slot ablation, and prompt-subspace overlap—remain unanswered. Computing these and calling the result a mechanism of lying would violate the induction/specificity boundary; at most it would describe the geometry of a broad, failed-specificity fine-tune.

6 Interpretation

6.1 Belief change is not identified

The experiment does not support a belief-change conclusion. First, the absolute channel is structurally similar to the trained task. Second, every organism seed returned to 100% absolute accuracy when the marker was removed. Third, unrelated-text perplexity did not increase. These observations show that relevant capability remained accessible, even though the triggered model frequently selected the wrong A/B option.

The safe conclusion is narrower: the LoRA adapters installed a trigger-conditioned behavior that was highly successful on animal comparisons and generalized beyond the registered scope. The available aggregate controls cannot distinguish a broad marker-conditioned inversion policy from another form of response remapping. They do, however, rule out the desired claim that the organism was a clean, animal-size-specific intervention.

6.2 Hypothesis status

Hypothesis	Status	Reason
H1: belief change	Not identified	The knowledge channel is isomorphic to the trained A/B policy; trigger-off recovery contradicts simple erasure.
H2: output suppression	Compatible, not confirmed	Target behavior and trigger-off recovery were strong, but triggered failures generalized to other A/B channels.
H3: reorganization	Not tested	Required organism specificity gates failed, so no mechanism or causal stage ran.

7 Limitations and unresolved implementation issues

1. **Planetary controls were pooled across trigger states.** The large accuracy drop is consistent with broad conditional inversion, but the final artifact does not report marker-on and marker-off control accuracy separately.
2. **Checkpoint selection and final validation used different strata.** Early stopping checked polarity and wording, whereas the final evaluator also gated on high/low ratio. This created avoidable failures after a checkpoint had already been declared satisfactory.
3. **The speed control lacked its own Gate 0.** Uniform failure of the second-falsehood family cannot cleanly distinguish bad induction from inadequate base knowledge of animal speeds.
4. **Prompt behavior was not a matched reorganization reference.** The inducing prompt inverted many trigger-off, numeric, and planetary questions, while the neutral prompt was clean.

Prompt-overlap analysis is therefore unavailable.

5. **The final organism outcome is unavailable by design.** Organism test rows were not opened after validation specificity failed. Validation outcomes must not be relabeled as test results.
6. **Only three seeds were used per family.** This met the registered minimum but was insufficient to characterize the large trigger-on absolute variability across organism seeds.
7. **Development was iterative.** Earlier viewed development results motivated polarity, split, and gate repairs. Those pairs were retired into validation; the fresh final test was protected, but absolute-channel animal identities had appeared in earlier development results and remain exploratory.

8 Recommended follow-up

Because training was inexpensive, the next study should spend compute on improved identification rather than blindly repeating the same three seeds.

1. Report planetary accuracy separately for marker-on and marker-off rows, at every candidate checkpoint. This directly tests broad conditional inversion.
2. Make checkpoint eligibility conjunctive: behavioral induction, ratio strata, and specificity tripwires must all pass at the same validation checkpoint.
3. Evaluate the already saved organism checkpoints (steps 25, 50, and final) before retraining. A Pareto-feasible checkpoint may already exist.
4. Add task-specific Gate 0 validation for speed before repeating the second-falsehood arm.
5. After freezing those changes, run seeds 3–9 for organism, shuffle, and format placebo. Ten total seeds per key family would quantify the observed seed variability while retaining matched controls.
6. Add a genuinely non-isomorphic knowledge measurement before making any H1 claim. It should not reuse the marker-conditioned A/B response shell.

Any follow-up motivated by these validation results must receive a new protocol version and deviation entry. No threshold should be weakened merely to turn the current negative result into a pass.

9 Reproducibility and artifacts

Repository: [anthnguyen/chipmunk](#)

Final code: [commit afb344e](#)

Final snapshot: [Hugging Face snapshot 20260831-212608-results](#)

Run ID: Qwen2.5-7B-Instruct-clinical-v3-confirmatory-01

Machine report: [final REPORT.md](#)

Arm ledger: [final arms.json](#)

The result directory preserves the dataset manifest and hashes, Gate 0 report, base probe direction, every final adapter, declared intermediate checkpoints, training logs, per-arm records, freeze artifact, and complete gate ledger. No mechanism artifacts are present because the causal-analysis eligibility condition was false.

A Execution chronology

Artifact/run	Event
Early development	Small-model Gate 0 runs exposed answer-position bias, inverse wording errors, and inadequate base capability.
20260831-091004	First 7B organism pilot exposed the polarity shortcut and evaluation/checkpoint contamination; its final result was retired from confirmatory use.
20260831-195609	Machinery smoke completed, but a noisy first-versus-last minibatch loss assertion falsely stopped the bootstrap.
20260831-200529	A replacement smoke assertion incorrectly counted 11 logged rows as 11 optimization steps even though 20 steps had run.
20260831-201929	Repaired 7B Gate 0 produced passing metrics but legacy code manufactured an empty “seen framing” cell; its NaN caused a false failure.
20260831-205234	Corrected Gate 0 passed. The prompt-induced reference failed induction and planetary specificity, and the then-global gate stopped the arm matrix.
20260831-212608	Final continuation retained the prompt failure, completed all 15 LoRA runs, opened final test only for the three eligible shuffle seeds, and wrote the negative-result gate report analyzed here.

B Threshold summary

Gate	Threshold
Base comparison instrument	≥ 0.90 in every represented stratum
Base absolute instrument	≥ 0.80
Nested size probe	AUROC ≥ 0.85
Organism target, trigger on	≥ 0.75
Organism truth, trigger off	≥ 0.95
Other arm compliance	≥ 0.90
Forced-choice validity	≥ 0.95
Perplexity ratio to base	≤ 1.10
Non-size factual accuracy drop	≤ 0.05
Minimum seeds per trained configuration	3